一、(40%)Consider two different implementations, M1 and M2, of the same instruction set. There are three classes of instructions (A, B, and C) in the instruction set. M1 has a clock rate of 80 MHz and M2 has a clock rate of 100 MHz. The average number of cycles for each instruction class and their frequencies (for a typical program) are as follows:

Instruction Class	M1 Cycles per Instruction Class	M2 Cycles per Instruction Class	Frequency
A	1	2	60%
B	2	3	30%
C	4	4	10%

(a) Calculate the average CPI for each machine, M1, and M2.
(b) Calculate the average MIPS ratings for each machine, M1 and M2.

(a) Calculate the average CPI for each machine, M1, and M2.
For Machine M1:
Cycles per Instruction = (60/100)* 1 + (30/100)*2 + (10/100)*4
= 1.6
For Machine M2:
Cycles per Instruction = (60/100)*2 + (30/100)*3 + (10/100)*4
= 2.5
(b) Calculate the average MIPS ratings for each machine, M1 and M2.
For Machine M1:
Average MIPS rating = Clock Rate/(CPI * 10_6)
= (80 * 10_6) / (1.6*10_6)
= 50.0
For Machine M2:
Average MIPS rating = Clock Rate/(CPI * 10_6)
= (100 * 10_6) / (2.5*10_6)
= 40.0

二、(20%)In the snippet of RISC-V assembler code below, how many times is instruction memory accessed? How many times is data memory accessed? (Count only accesses to memory, not registers.)

```
ld   x1, 0(x10)
addi   x10, x10, 1
sd   x1, 0(x10)
addi   x10, x10, 1
```

The instruction memory is accessed four times (as there are four instructions) and the data memory is
accessed twice (once for the `ld` instruction and another time for the `sd` instruction).

三、answer

1) Assume that integer is stored as a word (4 bytes) in memory,

and -2 is stored at Memory address start from 0x00000004, so what Byte is store in memory address 0x00000007 in big-endian computer? ( )

A. 11111110 B. 11111111 C. 10000000 D. 00000010

答案：A

```
11111111 11111111 11111111 11111110
4        5        6        7
```

2) What is the range of exponent of IEEE 754 single precision? ( )

A. 1~255 B. -128~127 C. 0~127 D. -126~+127

答案：D

3) Assume the current PC is 0x10000010, what's next value of PC after execution of "jal x0, 100"? ( )

A.0x00000064 B. 0x10000014 C. 0x10000074 D. 0x10000064

答案：C

```
100 = 0x64
Updated PC = current PC + offset = 0x10000010 + 0x64
= 0x10000074
```

4) What is the RISC-V assmbly code for the binary? ( )[p.119]

01000000101011110001000000110011

A. add x16, x15, x5 B. sub x16, x15, x5

C. add x16, x5, x15 D. sub x16, x5, x15

答案：B